

# ClubOps AI Event Operations Platform – Problem Analysis

## Problem Summary

College club event organizers currently manage tasks, volunteers, documents, meetings and announcements using fragmented tools (WhatsApp chats, spreadsheets, documents, personal notes). As clubs grow events, this scattered approach makes it hard to track deadlines, dependencies, action items and risks. The lack of a unified system causes confusion for organizers and volunteers, who may miss tasks or deadlines. A centralized AI-powered platform could streamline planning and coordination, improving event execution and reducing errors. In similar contexts, **digital tools have been shown to streamline volunteer management and event planning tasks** [47†L44-L50] , highlighting the importance of adopting an integrated solution.

## Core Problem

**“How might we help college club event organizers to seamlessly plan and execute events by addressing fragmented and manual coordination of tasks, volunteers, meetings, deadlines, documents and risks, while considering limited resources and the need for easy adoption?”**

## Problem Context

Currently, college clubs rely on ad-hoc tools and informal processes: WhatsApp groups for communication, Excel/Google Sheets for task lists, shared drives or Google Docs for documents, and handwritten meeting notes. This creates **siloed information and manual overhead**. For example, action items from meetings may only be conveyed via chat after the fact, and volunteers self-assign via spreadsheet without automated reminders or ownership tracking. As noted in volunteer management research, **digital platforms can enhance efficiency and trust in such contexts** [47†L44-L50] . However, existing generic tools (chat apps, spreadsheets) lack integration, so organizers repeatedly reconcile data manually. The current approach is insufficient because it does not scale: it is error-prone, time-consuming, and makes it hard to maintain a single “source of truth” for the event. This results in missed tasks, unclear responsibilities, and last-minute rushes that could compromise the event.

## Target Users / Stakeholders

- **Primary Users:**
  - *Club Event Organizers / Committee Members* – Responsible for planning, assigning tasks, scheduling meetings, and coordinating volunteers. They struggle with staying organized as events scale.
  - *Club Volunteers / Members* – Contribute to events (perform tasks, attend meetings) and need clarity on their assignments and deadlines. They often juggle tasks in addition to studies, so they need a clear view of responsibilities.

- **Secondary Users:**

- *Club Presidents / Administrators* – Oversee multiple events and care about overall club performance and resource allocation. They need reports and dashboards.
- *Faculty Advisors / Mentors* – Provide guidance and approvals (e.g., budgets, risk assessments) and should see high-level plans and risks.
- *Participants / Attendees* – The target audience for event announcements; their experience depends on successful execution (though they are not direct system users).

- **Organizations / Other Stakeholders:**

- *College or University Club Council* – May require oversight (approving events, budgets, safety). They may use reports or dashboards.
- *IT/Security Teams* – Concerned with data privacy and integration with campus systems (email, calendars).
- *Funding Sources / Sponsors* – Interested in event outcomes, but not direct users of the system.

## Pain Points

- **Fragmented Communication (Critical):** Information is scattered across WhatsApp, email, and personal notes. *Organizers* and *volunteers* often miss messages or tasks because there is no single task board. This leads to confusion and overlooked responsibilities.
- **Manual Task Tracking (Critical):** Tasks and deadlines are managed by hand in spreadsheets or chat. *Organizers* spend excessive time updating and reminding volunteers, and *volunteers* may miss tasks due to no automated reminders. This increases the chance that important tasks fall through the cracks.
- **Lack of Clear Ownership (Important):** There is no automated way to assign and remind task owners. *Volunteers* are unsure who is responsible for what until asked directly, and *organizers* must manually monitor progress. This ambiguity can slow progress and cause redundant work.
- **Meeting Notes Processing (Important):** After meetings, notes or transcripts are manually reviewed. Extracting action items and deadlines is tedious. *Organizers* struggle to convert discussions into a clear to-do list, so some action items may be missed.
- **Document Sprawl (Secondary):** Event documents (plans, designs, rules) are scattered in drives or chat. It's hard for anyone to find necessary documents quickly. This wastes time for *all stakeholders* and risks losing key information.
- **Risk Management Gap (Secondary):** Potential risks (budget overruns, safety issues, weather impacts) are not systematically identified. *Organizers* rely on experience or checklists. Unexpected problems may catch the club unprepared.
- **Manual Announcements (Secondary):** Event announcements and communications are handcrafted. *Organizers* spend time writing emails or posts, and *members* may miss announcements without multi-channel delivery.

## Root Cause Analysis

- **Direct Causes:** Reliance on disparate manual tools (WhatsApp, spreadsheets, docs) for critical event tasks. Lack of a unified platform leads to fragmentation. Inadequate processes for extracting and assigning tasks from meetings.
- **Contributing Factors:** Small club budgets and volunteer-managed culture discourage investment in specialized software. Users may lack training or habit for structured project management, preferring

familiar apps. There are no existing standards adopted across clubs, so each event is managed ad-hoc.

- **Root Causes:** The absence of a centralized event management system with automation. Clubs have not adopted event-specific management tools; thus information and processes remain siloed. (Root cause cannot be conclusively determined from the provided information beyond the lack of integration and automation.)

## Current Process

1. **Task Assignment:** Club officers create task lists in spreadsheets or chat. They share via Google Drive or WhatsApp.
2. **Volunteer Sign-Up:** Volunteers join a WhatsApp group or email thread, and informally claim tasks or are asked by officers. No formal system to track assignments.
3. **Meetings:** Meetings occur (in person or virtual). Minutes are taken as text or voice recording.
4. **Post-Meeting Follow-up:** Someone manually reviews notes or recordings, types up action items in a doc or chat, and notifies individuals. This can lead to delays or omissions.
5. **Document Management:** Plans and references are saved in shared drives or shared chats, without a structured repository. Finding documents requires manual search.
6. **Deadlines and Reminders:** Deadlines are set in the spreadsheet or personal calendars. Reminders are manual (via chat ping or email), not automated.
7. **Risk Oversight:** Risks are informally discussed, often offhand, with no centralized register or alerts.
8. **Announcements:** Event announcements are composed manually and posted on group chat or social media.

**Failure Points:** The handoff between these steps is manual. For example, after a meeting (Step 4), if someone doesn't immediately publish action items, tasks may be forgotten. Using general-purpose chat means important updates can be buried. There is no automatic check for overlapping deadlines or missing assignees, so scheduling conflicts are common.

## Existing Solutions / Current Alternatives

- **Generic Tools (Spreadsheets, Chats, Drive):** Clubs use common apps (Google Sheets, WhatsApp, Discord, Email, Google Drive) to coordinate. These require manual updating and do not integrate. *Limitation:* They solve only one piece of the puzzle (e.g. a group chat or file sharing) and lack automation or insights. Important data (tasks, deadlines) must be tracked by hand, leading to errors and inefficiency.
- **Volunteer Mgmt Software:** Commercial products (e.g., Rosterfy, CERVIS) offer volunteer scheduling and event tools. *Limitation:* These may be overkill or too expensive for small clubs, and are not tailored for end-to-end event planning. They often focus on scheduling volunteers for shifts, not on planning tasks, meetings, or documents. Integration with club-specific workflows is limited.
- **Project/Task Management Tools:** Apps like Trello, Asana, or Microsoft Teams could be (informally) adopted to list tasks and deadlines. *Limitation:* Clubs rarely adopt them institutionally; they have a learning curve and no AI assistance. They also lack built-in meeting transcription or risk analysis.
- **Meeting Transcription Services:** Tools like Otter.ai can transcribe meetings. *Limitation:* While transcripts are useful, volunteers still must manually read and extract tasks. These tools do not automatically assign tasks or update calendars.

- **Wiki/Knowledge Base:** Some clubs use a wiki (e.g. Notion, Google Sites) for documentation.  
*Limitation:* These provide document storage but no dynamic assistance (no auto-summarization or task extraction).

None of these existing alternatives fully solve the integrated coordination problem. The proposed AI platform is meant to address these gaps by combining features.

## Desired Outcome

- **Integrated Workflow:** All event-related activities (task tracking, volunteer management, meetings, docs, risks) occur in one platform, reducing context-switching.
- **Efficiency and Accuracy:** Planning and coordination become faster and less error-prone. Organizers and volunteers spend less time on manual admin and more on execution.
- **Clarity of Roles and Deadlines:** Tasks are clearly listed with owners and deadlines. Automated reminders and status updates ensure tasks are completed on time.
- **Intelligent Assistance:** AI summarizes meetings, extracts action items and deadlines automatically, identifies owners (based on context or history), and flags risks (e.g. conflicting deadlines or resource shortages).
- **Accessible Knowledge Repository:** A searchable repository of club documents and past events (knowledge base) that the AI can use to answer questions or provide context (retrieval-augmented QA).
- **Improved Communication:** AI-generated announcements (e.g. event reminders, schedule updates) saved effort and ensure consistency. Volunteers receive clear, timely communications.

**Who benefits:** Club organizers (time saved, fewer mistakes), volunteers (clearer tasks, less confusion), and ultimately club members and attendees (better-run events).

**Measurable outcomes:** Reduced task completion delays (e.g. >X% tasks on time), higher volunteer satisfaction, fewer scheduling conflicts, time saved per event, improved attendance at events.

## Constraints

- **Technology Stack (given):** Likely must use suggested AI models/APIs (Gemini API, OpenAI API, etc.), plus React/Next.js front end, FastAPI/Python backend, MongoDB/PostgreSQL. We cannot assume undisclosed technologies or proprietary club systems.
- **Resources:** Typical college club has limited budget and technical resources. The solution should not require expensive infrastructure. (Implied by hackathon context.)
- **Scope/Complexity:** For an MVP/hackathon, focus on core features. Very complex workflows (e.g. full financial accounting, advanced analytics) are *out of scope*.
- **User Skill Level:** Club members are likely students with average tech savviness. The interface must be intuitive (no steep learning curve).
- **Privacy:** Club events may involve student personal data (names, roles). The system must respect privacy (e.g., not expose private info without permission).
- **Connectivity:** Likely web-based with internet (college setting). Should handle moderate concurrent users but not enterprise scale.
- **Regulations:** Probably no special regulations (not health or finance domain), but if any student data laws apply (e.g., GDPR for EU students), compliance must be considered (in privacy constraints).

## Data Requirements

- **User/Volunteer Profiles:** Data such as names, roles, contact info, skills/availability. *Source:* Club membership lists, sign-up forms. *Purpose:* Assign tasks and send notifications. *Availability:* Provided or collected (club likely has this data in membership roster).
- **Task Data:** Titles, descriptions, owners, deadlines, status. *Source:* Generated within platform or imported from existing spreadsheets. *Purpose:* Core of task management. *Availability:* New data, input by users or AI.
- **Meeting Transcripts/Notes:** Text or audio transcripts of meetings. *Source:* Recorded meetings (via integrated voice or manual notes). *Purpose:* To extract action items and decisions. *Availability:* Requires recording/collection during meetings.
- **Documents/Knowledge Base:** Event plans, past reports, guidelines, venue info. *Source:* Club drives, shared documents, public info. *Purpose:* RAG for answering questions, context for decisions. *Availability:* Likely available from club archives (requires import).
- **Calendar/Deadlines:** Event dates, submission deadlines. *Source:* Input by users or extracted from tasks. *Purpose:* Scheduling and reminders. *Availability:* Derived from task and event data.
- **Risk/Issue Data:** Possibly risk checklists or incident reports. *Source:* User input and meeting discussions. *Purpose:* AI to flag and explain potential risks. *Availability:* Requires capturing from discussions and past events.
- **Communication Logs:** Announcements, emails, chat logs. *Source:* History of club communications. *Purpose:* Context, possibly training AI to match club tone. *Availability:* May not be provided for privacy reasons.

## AI / ML Opportunities

- **Retrieval-Augmented Generation (RAG):** Use RAG to ground AI in club-specific documents and past event knowledge. For example, answering questions about policies or referencing previous events by retrieving relevant docs [38†L31-L39] . This ensures accuracy by using the club's own data.
- **Meeting Transcription & Summarization:** Apply ASR (speech-to-text) to record meetings (e.g., Gemini 3.5 Transcribe [3†L174-L178] ) and then NLP to extract summary of key points and action items. This automates step from meeting to tasks.
- **Action Item Extraction:** Use NLP models (like GPT or Claude) to parse transcripts or notes and generate a structured list of tasks, owners and deadlines. With **structured outputs**, models can return JSON formatted lists of tasks [51†L184-L188] , enabling direct injection into the task system.
- **Task Assignment & Recommendation:** Classification or recommendation models could suggest task owners based on keywords (e.g., if "poster design" was discussed, suggest the club's designer volunteer) or send reminders automatically.
- **Risk Identification:** Use text analysis on planning text or past incidents to predict or highlight risks (e.g., "If venue capacity is small and attendance is high, flag as risk"). This could use simple rules or ML classification on historical data.
- **Conversational AI / Chatbot:** A chat interface where organizers ask the AI (backed by GPT/Gemini) for summaries ("What are the pending tasks for Event X?") or it could proactively suggest next steps.
- **Generative Communications:** AI can draft announcements, meeting summaries or email reminders in natural language, ensuring clarity and saving time.
- **Non-AI (Rule-Based) Opportunities:** Some parts, such as core task tracking and volunteer sign-up, can be handled with standard app logic (databases, CRUD). Traditional alerts (calendar reminders) may not require AI.

AI is useful because the tasks involve understanding and generating language (meeting notes, planning text) and managing large unstructured data (docs, transcripts). For example, Gemini's structured output capability [51†L184-L188] enables automatic extraction of tasks. RAG improves information accuracy by retrieving club docs [38†L31-L39]. Conversely, straightforward tracking (storing tasks, user auth, UI rendering) can be done with normal backend/frontend code.

## Problem Scope

- **In Scope:**

- Centralized task and volunteer management (creating tasks, assigning volunteers, tracking status).
- Meeting note/transcript processing (capture, summary, action extraction).
- Document repository (uploading and searching event-related docs).
- Automated announcements and reminders (e.g. via email or in-app).
- AI features: action-item extraction, RAG-based Q&A, risk flagging for obvious issues.
- Basic user authentication and club membership management.
- A dashboard of tasks, deadlines, and upcoming events.

- **Out of Scope:**

- Financial management beyond basic cost tracking.
- Detailed venue logistics or ticketing systems.
- Integrations with external calendar (except perhaps basic export).
- Full general-purpose chat integrations (although optional notifications via WhatsApp may be considered later).
- Elaborate UI themes or non-event-related functions.
- Advanced security compliance beyond standard practices.

For a hackathon MVP, focus on core event/event lifecycle aspects: task/volunteer panel, basic meeting transcription demo, simple risk check, and a small knowledge base. Advanced features (multi-agent workflows, extensive analytics) are deferred.

## MVP Definition

- **Must-Have Functionality:**

- **User Login/Profile:** Club officers and volunteers can log in.
- **Task Management:** Create tasks with title/description; assign to volunteers; set deadlines and status. Display tasks in a dashboard.
- **Volunteer Management:** List of club volunteers; ability to select a volunteer as task owner.
- **Meeting Input:** Upload meeting notes or audio; AI transcribes (simulated) and lists tasks with owners/dates (structured output).
- **Document Repository:** Ability to upload and search relevant documents. (At least show a list with search.)
- **Announcements Module:** Compose or auto-generate an announcement for an event; preview it.
- **Basic AI Integration:** Use one LLM API (e.g., Gemini or ChatGPT) to process meeting text for action items and to answer questions from a small knowledge base (demo RAG).

- **UI/Frontend:** A simple React/Next.js interface with pages for dashboard, tasks, meetings, docs, and announcements.
- **Backend/Database:** FastAPI service with endpoints for tasks, meetings, docs; MongoDB or PostgreSQL to store data.
- **Demo Flow:** Show a sample event where tasks are created manually, then a meeting transcript is processed to auto-add tasks, and an announcement is drafted by AI.
- **Essential User Flow:** An organizer logs in → views event dashboard → adds a task or uploads meeting notes → AI extracts tasks, which appear in the task list → AI flags a potential risk or suggests a deadline → organizer reviews and sends an announcement drafted by AI to volunteers.
- **Minimum Required Data:** A small example club dataset (e.g., 1 event with 3 volunteers, a short meeting transcript, 2-3 sample documents).
- **Minimum AI Capability:** At least one model call demonstrating text generation or summarization (e.g., extraction of action items into JSON and one RAG query from the document repo). If Gemini/ChatGPT calls are available, use one to extract tasks.
- **Expected Demo Outcome:** A live or recorded demo where the audience sees: tasks being created, an upload of meeting text triggering AI-generated tasks with deadlines, and an AI-generated announcement message, all within the interface.

## Success Metrics

- **Task Completion Rate:** Measure the percentage of tasks completed by their deadlines before vs. after using the system. (Aim for >90% on time.)
- **Time Saved:** Estimate reduction in manual effort. E.g., time to summarize meeting or create announcements (could be time-stamped actions before/after).
- **System Usage:** Adoption rate – number of tasks or actions managed in the system per event. Higher numbers indicate usage.
- **Accuracy of AI Extraction:** Percentage of correctly identified action items from meeting notes (compared to a human-annotated list). E.g., aim >80% precision/recall.
- **User Satisfaction:** Survey organizers on ease-of-use before/after (Likert scale); or track number of help requests.
- **Error Reduction:** Compare number of missed tasks/overlaps before vs. after (if historical data exists).

Each metric could be measured by logging system events (task completions), manually auditing AI output, and surveying users after a trial period.

## Risks and Challenges

- **Data Quality:** Meeting transcripts may be noisy (speaker overlap, slang), causing poor extraction accuracy. *Mitigation:* Use clear templates for meeting minutes; allow manual edits of AI output before commit. Possibly fallback to a simpler summarization if transcription is bad.

- **Limited Training Data:** Club-specific terminology may not be in model knowledge, causing inaccuracies. *Mitigation:* Use RAG with club documents to ground answers; fine-tune prompts with known event terminology.
- **AI Hallucination:** LLM might add tasks or risks that are unfounded. *Mitigation:* Use structured output constraints (e.g. JSON format) 【51†L184-L188】 and double-check AI suggestions before applying. Provide a “review” step for users.
- **API Dependency & Cost:** Reliance on external AI APIs (Gemini/OpenAI) may incur latency or costs, especially at scale. *Mitigation:* Cache frequent queries, use cheaper model variants when possible. Ensure fallback or offline mode (basic UI) if AI call fails.
- **Scalability:** If many tasks and docs accumulate, performance may degrade. *Mitigation:* Use efficient database indexing; restrict demo data size. For production, consider paging and search indexing.
- **Privacy/Security:** Storing meeting transcripts and volunteer info could raise privacy concerns. *Mitigation:* Ensure data is encrypted at rest/in transit, require authentication (maybe SSO via college), and allow users to control what info is saved (e.g., anonymize transcripts).
- **User Resistance:** Volunteers might resist adopting a new system if it's too complex. *Mitigation:* Keep UI simple and mobile-friendly. Possibly integrate familiar channels (like sending Slack/WhatsApp updates from the app). Offer onboarding/tutorial.

## Assumptions

- **ASSUMPTION-01:** Volunteers and club members have internet access and are comfortable using a web app (not limited to low-tech solutions).
- **ASSUMPTION-02:** The club has existing digital data (documents, membership list) that can be loaded; minimal manual data entry.
- **ASSUMPTION-03:** AI APIs (Gemini/OpenAI) are available and permitted for use in the project (budget/time for API calls).
- **ASSUMPTION-04:** Events are of moderate complexity (not multi-million attendee), appropriate for a student-run club scale.
- **ASSUMPTION-05:** Security requirements align with standard web best practices (not needing special medical/financial compliance).

## Ambiguities / Questions

- **Q1. Is this platform intended for one club or multiple clubs?**  
*Why it matters:* If multi-club support is needed, the system must handle different user groups, permissions, and data segregation. It affects user management design.
- **Q2. What communication channels do clubs currently prioritize?** (e.g., WhatsApp vs. email vs. Slack)  
*Why it matters:* Integration or notification preferences will change design choices for announcements and reminders. If volunteers primarily use WhatsApp, we may need a bot; if email, then SMTP integration.
- **Q3. What data privacy policies apply to student information?**  
*Why it matters:* Handling personal data (names, emails, meeting audio) may trigger institutional policies or regulations (e.g., FERPA/GDPR). We must know limits for data storage and sharing.

- **Q4. How are risks currently identified and managed?**

*Why it matters:* Understanding existing risk processes helps determine the kind of risk analysis or checklists the AI should use. It clarifies if “risk identification” means safety, budget, schedule, or all.

- **Q5. What are the typical sizes of events and teams?**

*Why it matters:* The scale (number of tasks, volunteers) affects required system capacity and feature scope. A small club event has different needs than a large intercollegiate fest.

- **Q6. Are there existing tools or spreadsheets in use?**

*Why it matters:* If clubs already use certain formats (e.g., a specific spreadsheet template), we may import data or ensure compatibility, rather than reinvent those formats.

## Final Formulated Problem

**Problem Title:** AI-Powered Club Event Management

**Target Users:** College student club event organizers and volunteers

**Current Situation:** Clubs manage event tasks, meetings, documents, and communication using disconnected tools (WhatsApp, spreadsheets, separate notes), leading to lost information and inefficiency [47†L44-L50] .

**Core Problem:** Event coordination is fragmented and manual, making it difficult to track responsibilities, deadlines, and insights across tasks, volunteers, meetings, and documents.

**Root Cause:** Lack of a centralized, automated platform for event planning and execution; dependence on ad-hoc tools forces manual coordination of interconnected information.

**Impact:** Important tasks or deadlines may be missed, volunteer workloads overlap or go unseen, and organizers spend excessive effort on administration instead of execution.

**Desired Outcome:** A unified platform that consolidates event tasks, volunteer roles, meetings, docs, and risks into one system with AI assistance. This would provide automatic transcription and task extraction, clear assignments and alerts, and an accessible knowledge repository, enabling smoother, more reliable event planning.

**Key Constraints:** Must use web stack (React/Next.js, FastAPI/Python), available AI APIs (Gemini, OpenAI), and be feasible for a student-team hackathon timeline. Focus on core event lifecycle features; integration with existing communication channels is optional.

**MVP Scope:** Login system; dashboard for tasks and volunteers; ability to upload meeting text to generate action items via an LLM; simple document repository search; AI-generated announcement draft; basic user flows for creating and assigning tasks, and reviewing AI suggestions.

## One-Line Problem Definition

"College club organizers struggle with coordinating event planning tasks and information across fragmented tools, resulting in missed deadlines and confusion, because they lack a centralized, intelligent platform, creating a need for an AI-powered system that unifies tasks, communication, and knowledge for smooth event execution."

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